

## WICHITA FALLS SHEPPARD AFB, TX, USA

WMO: 723510

Lat: 33.979N

Lon: 98.493W

Elev: 1017

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13966

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 18.6       | 22.5 | 4.6                           | 7.2 | 25.1 | 9.0 | 9.1 | 31.2 | 31.4                  | 43.1 | 27.8 | 46.1 | 10.5                  | 10   | 0.582 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.0                   | 103.8           | 72.7 | 101.2 | 72.9 | 98.5 | 73.1 | 77.8                | 92.1 | 76.8 | 91.3 | 75.7 | 90.2 | 12.2                 | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.0                            | 131.6 | 83.3 | 72.8 | 126.5 | 82.1 | 71.9 | 122.5 | 81.4 | 41.9          | 92.4 | 40.8 | 91.1 | 39.8 | 90.3 | 88.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 27.3              | 24.7 | 22.0 | 11.9                       | 107.2 | 3.8                | 3.3 | 9.1                                                | 109.5 | 6.9        | 111.5 | 4.7        | 113.3 | 2.0        | 115.7 |      |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |      |
|                   |      |      | WB                         | 10.2  | 80.5               | 3.6 | 2.6                                                | 7.6   | 82.4       | 5.5   | 83.9       | 3.5   | 85.4       | 0.9   | 87.3 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 64.4   | 43.1  | 46.8  | 55.2  | 63.4  | 72.5  | 80.8  | 85.2  | 84.8  | 76.9  | 65.4  | 53.8  | 44.1 |
|                                                                               | DBStd       | 17.21  | 10.16 | 10.99 | 10.31 | 8.31  | 7.67  | 5.28  | 4.92  | 5.18  | 6.95  | 8.65  | 9.33  | 9.51 |
|                                                                               | HDD50       | 806    | 259   | 172   | 65    | 8     | 0     | 0     | 0     | 0     | 0     | 6     | 65    | 231  |
|                                                                               | HDD65       | 2774   | 678   | 513   | 329   | 125   | 24    | 0     | 0     | 0     | 6     | 102   | 348   | 649  |
|                                                                               | CDD50       | 6079   | 47    | 83    | 226   | 410   | 699   | 924   | 1091  | 1080  | 808   | 483   | 180   | 48   |
|                                                                               | CDD65       | 2574   | 1     | 4     | 26    | 77    | 258   | 474   | 626   | 615   | 365   | 114   | 13    | 1    |
|                                                                               | CDH74       | 31464  | 23    | 74    | 324   | 858   | 2732  | 5664  | 8468  | 8079  | 3938  | 1144  | 141   | 19   |
|                                                                               | CDH80       | 16819  | 2     | 16    | 85    | 266   | 1204  | 3006  | 5045  | 4786  | 1975  | 406   | 26    | 2    |
| Wind                                                                          | WSAvg       | 11.3   | 11.1  | 11.6  | 12.5  | 13.2  | 12.2  | 11.8  | 10.6  | 9.8   | 9.7   | 10.8  | 11.2  | 10.7 |
| Precipitation                                                                 | PrecAvg     | 28.6   | 1.1   | 1.5   | 2.0   | 2.8   | 3.8   | 3.3   | 1.8   | 2.6   | 3.4   | 2.9   | 1.6   | 1.5  |
|                                                                               | PrecMax     | 47.4   | 4.5   | 4.6   | 6.3   | 8.0   | 17.0  | 8.6   | 5.6   | 8.3   | 10.2  | 8.4   | 6.9   | 6.9  |
|                                                                               | PrecMin     | 14.3   | 0.0   | 0.0   | 0.1   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                               | PrecStd     | 6.4    | 0.9   | 1.1   | 1.4   | 1.9   | 3.1   | 2.3   | 1.5   | 1.9   | 2.5   | 2.2   | 1.5   | 1.4  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 77.4  | 81.5  | 87.4  | 92.2  | 99.6  | 104.4 | 107.3 | 107.3 | 102.6 | 93.2  | 83.7  | 76.5 |
|                                                                               |             | MCWB   | 56.9  | 59.0  | 61.7  | 64.8  | 72.5  | 72.7  | 72.1  | 72.0  | 71.4  | 68.5  | 64.8  | 57.7 |
|                                                                               | 2%          | DB     | 70.6  | 75.3  | 81.7  | 86.6  | 94.7  | 99.8  | 103.7 | 103.7 | 97.6  | 88.8  | 78.1  | 69.8 |
|                                                                               |             | MCWB   | 54.2  | 57.2  | 62.1  | 65.4  | 71.6  | 72.6  | 72.7  | 72.8  | 72.1  | 68.9  | 61.9  | 55.4 |
|                                                                               | 5%          | DB     | 65.4  | 70.1  | 77.3  | 82.6  | 90.8  | 96.4  | 101.0 | 101.0 | 94.1  | 84.6  | 73.7  | 64.8 |
|                                                                               |             | MCWB   | 51.2  | 55.3  | 60.2  | 64.3  | 70.9  | 73.0  | 73.1  | 73.3  | 71.8  | 67.0  | 59.8  | 53.1 |
|                                                                               | 10%         | DB     | 60.4  | 64.9  | 72.5  | 79.0  | 86.8  | 93.3  | 98.1  | 98.1  | 90.7  | 80.6  | 69.4  | 60.0 |
|                                                                               |             | MCWB   | 49.2  | 52.8  | 58.3  | 63.3  | 70.1  | 73.6  | 73.6  | 73.4  | 71.2  | 65.5  | 58.6  | 50.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 62.4  | 64.3  | 68.3  | 72.8  | 77.4  | 79.6  | 79.4  | 78.4  | 77.1  | 74.3  | 68.9  | 64.0 |
|                                                                               |             | MCDB   | 69.6  | 72.8  | 80.0  | 82.1  | 89.6  | 91.8  | 93.5  | 94.3  | 90.7  | 85.1  | 77.1  | 70.5 |
|                                                                               | 2%          | WB     | 57.8  | 61.4  | 65.7  | 70.2  | 75.2  | 77.5  | 77.6  | 77.2  | 75.4  | 72.3  | 65.3  | 59.1 |
|                                                                               |             | MCDB   | 65.2  | 69.2  | 75.3  | 79.7  | 88.1  | 90.6  | 92.5  | 92.9  | 88.3  | 82.5  | 73.9  | 64.9 |
|                                                                               | 5%          | WB     | 53.3  | 58.3  | 63.5  | 68.3  | 73.7  | 76.2  | 76.6  | 76.1  | 74.2  | 70.4  | 63.0  | 55.0 |
|                                                                               |             | MCDB   | 62.6  | 67.3  | 72.5  | 77.5  | 85.9  | 89.4  | 91.6  | 91.7  | 86.9  | 79.3  | 70.5  | 62.6 |
|                                                                               | 10%         | WB     | 49.7  | 54.1  | 60.8  | 66.3  | 72.1  | 75.0  | 75.4  | 75.1  | 73.1  | 68.5  | 60.4  | 50.5 |
|                                                                               |             | MCDB   | 59.9  | 64.5  | 70.1  | 75.2  | 83.4  | 88.2  | 90.4  | 90.5  | 85.3  | 76.9  | 67.8  | 59.4 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 23.0  | 23.3  | 24.1  | 24.0  | 22.2  | 21.7  | 23.0  | 23.1  | 22.5  | 23.1  | 22.9  | 21.9 |
|                                                                               |             | MCDBR  | 31.3  | 31.6  | 30.5  | 28.9  | 26.7  | 25.9  | 26.2  | 26.4  | 26.3  | 27.4  | 28.1  | 29.7 |
|                                                                               |             | MCWBR  | 17.6  | 17.1  | 15.3  | 14.1  | 10.5  | 7.7   | 6.7   | 7.2   | 8.0   | 12.2  | 15.1  | 17.5 |
|                                                                               | 5% WB       | MCDBR  | 27.8  | 28.3  | 24.5  | 23.1  | 23.0  | 21.5  | 22.2  | 22.6  | 21.3  | 21.2  | 23.5  | 24.6 |
| MCWBR                                                                         |             | 17.6   | 18.6  | 13.9  | 13.5  | 11.1  | 8.9   | 7.9   | 7.2   | 7.8   | 11.2  | 15.5  | 17.2  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.293  | 0.304 | 0.337 | 0.373 | 0.411 | 0.427 | 0.426 | 0.419 | 0.387 | 0.352 | 0.317 | 0.297 |      |
|                                                                               | taud        | 2.518  | 2.501 | 2.422 | 2.349 | 2.312 | 2.303 | 2.345 | 2.311 | 2.376 | 2.459 | 2.492 | 2.506 |      |
|                                                                               | Ebn at Noon | 294    | 302   | 298   | 290   | 278   | 272   | 272   | 272   | 277   | 280   | 282   | 285   |      |
|                                                                               | Edh at Noon | 26     | 30    | 35    | 39    | 41    | 42    | 40    | 40    | 36    | 31    | 27    | 25    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 951    | 1138  | 1514  | 1840  | 1974  | 2184  | 2188  | 2004  | 1666  | 1294  | 1003  | 816   |      |
|                                                                               | RadStd      | 97     | 166   | 94    | 149   | 206   | 188   | 136   | 116   | 139   | 185   | 126   | 93    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (41 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page